Classification of Foetal Heart Rate

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1 Introduction

Biostatistics (also known as **biometry**) is a branch of statistics that applies statistical methods to a wide range of topics in biology. And one topic I will be looking into is **"Classification of Foetal Heart Rate"**.
Cardiotocography involves observing the fetal heart rate (FHR) in conjunction with uterine contractions. The connection between these two factors is generally recognized as an indicator of the fetus's oxygenation levels. This process aims to support both the mother and the baby by offering obstetricians valuable, real-time data that can guide decisions about the need for medical intervention (Kauffmann and Silberman, 2023). Understanding fetal heart rate (FHR) tracings has been challenging due to inconsistencies in terminology and definitions. Research indicates that the interpretation of these tracings often lacks reliability (Hooper and Elsamadicy, 2014). The goal of this study is to create a classification model with the highest possible precision to support gynecologists in their embryology diagnoses.

2 Data

2.1 Dataset

The raw dataset consists of 2126 fetal cardiotocograms (CTGs) with 22 explanatory variables and 2 target variables (CLASS, NSP) all described in Table 1 with no missing values (Bache et al.,2010).

Variable	Description
`b`	Start time of the CTG
`e`	End time
`LBE` / `LB`	FHR baseline (beats per minute)
`AC`	Number of accelerations
`FM`	Number of fetal movements
`UC`	Number of uterine contractions
`DL`	Number of light decelerations
`DS`	Number of severe decelerations
`DP`	Number of prolonged decelerations
`DR`	Number of repetitive decelerations
`ASTV`	% of time with abnormal short-term variability
`MSTV`	Mean short-term variability
`ALTV`	% of time with abnormal long-term variability
`MLTV`	Mean long-term variability
`Width`	Width of FHR histogram
`Min`	Minimum of FHR histogram
`Max`	Maximum of FHR histogram
`Nmax`	Number of histogram peaks
`Nzeros`	Number of histogram zeros
`Mode`	Histogram mode
`Mean`	Histogram mean
`Median`	Histogram median
`Variance`	Histogram variance
`Tendency`	Histogram tendency
`CLASS`	FHR pattern class code (1 to 10)
`NSP`	Fetal state class (N = normal; S = suspect; P = pathologic)

Table 1: Description of Variables in the CTG Dataset

2.2 Data Preprocessing

- **New Variables** AC, FM, UC, DL, DS, DP, and DR are count based variables that naturally depend on the duration of the CTG recording. AC1, FM1, UC1, DL1, DS1, DP1, and DR1 are the new variables created in Excel where these variables are expressing the original counts as

rates per second. For example,

$$\text{AC1} = \frac{\text{AC}}{e - b}$$

and similarly for the others.

- **Normalization** To ensure all variables contribute equally to model training and to mitigate the effects of differing measurement scales, Min-Max normalization was applied to all numerical features in R. This technique rescales each variable to a common range $[0, 1]$, using the formula:

$$x_{\text{normalized}} = \frac{x - \min(x)}{\max(x) - \min(x)}$$

This is especially important when working with variables of varying units and magnitudes, as in this dataset, which contains a mix of counts, percentages, and histogram-based statistics. By applying Min-Max normalization, it ensures that the learning algorithms are not biased toward features with larger numeric ranges.

2.3 Response variables

The two response variables are:

- **NSP (Normal, Suspect, Pathologic)**: The NSP classification involves three distinct classes:

$$\text{NSP} = \{\text{Normal}, \text{Suspect}, \text{Pathologic}\}.$$

- **Simplicity:** The NSP variable is a simpler target for the classification model, as it only requires distinguishing between three classes. This simplicity allows for easier model training and reduces the complexity of classification tasks.

- **CLASS (1-10)**:The CLASS variable involves more detailed fetal heart rate classifications:

$$\text{CLASS} = \{1, 2, 3, \ldots, 10\}.$$

- **Complexity:** The CLASS variable has 10 classes, each corresponding to a more granular distinction in fetal heart rate patterns. This requires the model to capture subtle differences in the data, leading to more complex classification boundaries.
"Based on the histogram plots, Figures 1 and 2, the NSP variable (with three classes) demonstrates clearer class separation, making it more robust to small variations and likely better suited for simpler models. In contrast, the CLASS variable (with 10 categories) provides more detailed information but introduces greater complexity and potential overlap between classes, which may require advanced techniques to accurately classify."

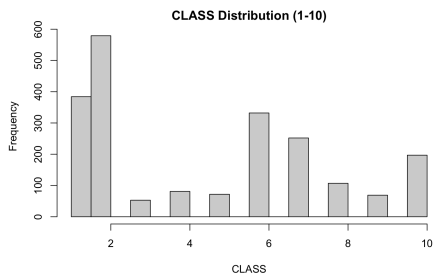

Figure 1: Higher complexity with 10 categories, requiring more nuanced discrimination. May suffer from lower accuracy if the model struggles with subtle differences.

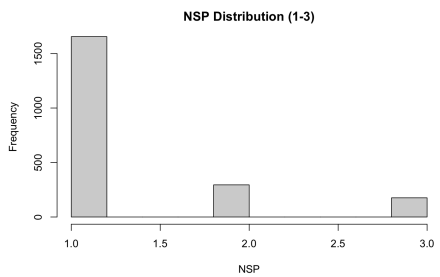

Figure 2: Simpler with only 3 categories, making it easier to classify. Likely more robust to minor variations in features.

3 Methodology

3.1 Linear Discriminant Analysis (LDA)

Initially, Linear Discriminant Analysis (LDA) was considered due to its suitability for multi-class classification, specifically for the Normal, Suspicious, and Pathological fetal states. LDA's strength lies in its ability to reduce the dimensionality of the data while aiming to maximize the separation between these classes, potentially simplifying the analysis of features such as Fetal Heart Rate (FHR) and its variability. However, the subsequent reliance on histograms in Figure 3 for data visualization revealed that many of the underlying variables did not exhibit a normal distribution, which is an assumption of LDA. Furthermore, the histograms suggest that several features are more discrete than continuous and display significant skewness to the left or right. LDA is not the suitable method for accurate classification (Fergus et al., 2017).

3.2 Binary Logistic Regression (BLR)

The categorically distinct distributions observed in the histograms of Figure 3 suggest that logistic regression is an appropriate modelling approach. Logistic Regression (LR) is a type of non-linear regression used to model the probability of a binary response variable based on a set of independent predictor variables (Abdulqader, 2017). Introduced by Cox in 1958, LR applies the logistic (sigmoid) function to a linear combination of input features to estimate the probability of an event occurring(Afridi et al., 2019). This makes it well-suited for classification problems involving two outcome categories.

3.2.1 Kruskal-Wallis test

The Kruskal-Wallis test is performed before modelling BLR as part of the exploratory data analysis. Although it does not assume normality, the test is specifically designed for comparing continuous or ordinal variables across groups defined by a categorical variable (Gobo et al., 2006). In our analysis, the predictor variables are continuous or ordinal, and the outcome variable (NSP) is categorical with three levels. Therefore, the Kruskal-Wallis test is an appropriate method to assess whether the distributions (particularly the medians) of the predictor variables differ significantly across the NSP classes. This helps identify potentially informative features for subsequent modelling.
The Kruskal-Wallis test statistic H is calculated as:

$$H = \frac{12}{n(n+1)} \sum_{i=1}^{k} \frac{R_i^2}{n_i} - 3(n+1)$$

where:

- n is the total number of observations across all groups,
- k is the number of groups,
- R_i is the sum of ranks for group i,
- n_i is the number of observations in group i.

Under the null hypothesis, H approximately follows a chi-squared distribution with $k-1$ degrees of freedom for sufficiently large sample sizes (Gobo et al., 2006).
The Kruskal-Wallis test was conducted in R to assess whether statistically significant differences exist in the distributions of various fetal heart rate (FHR) related features across the three fetal states (Normal, Suspect, and Pathologic) represented by the NSP variable. The results, summarized in Table 3.2.1, indicate that most of the features exhibit significant differences across these groups.

The Kruskal-Wallis test was conducted to evaluate differences in fetal heart rate (FHR) characteristics across three clinical classifications of fetal state (Normal, Suspect, and Pathologic). As shown

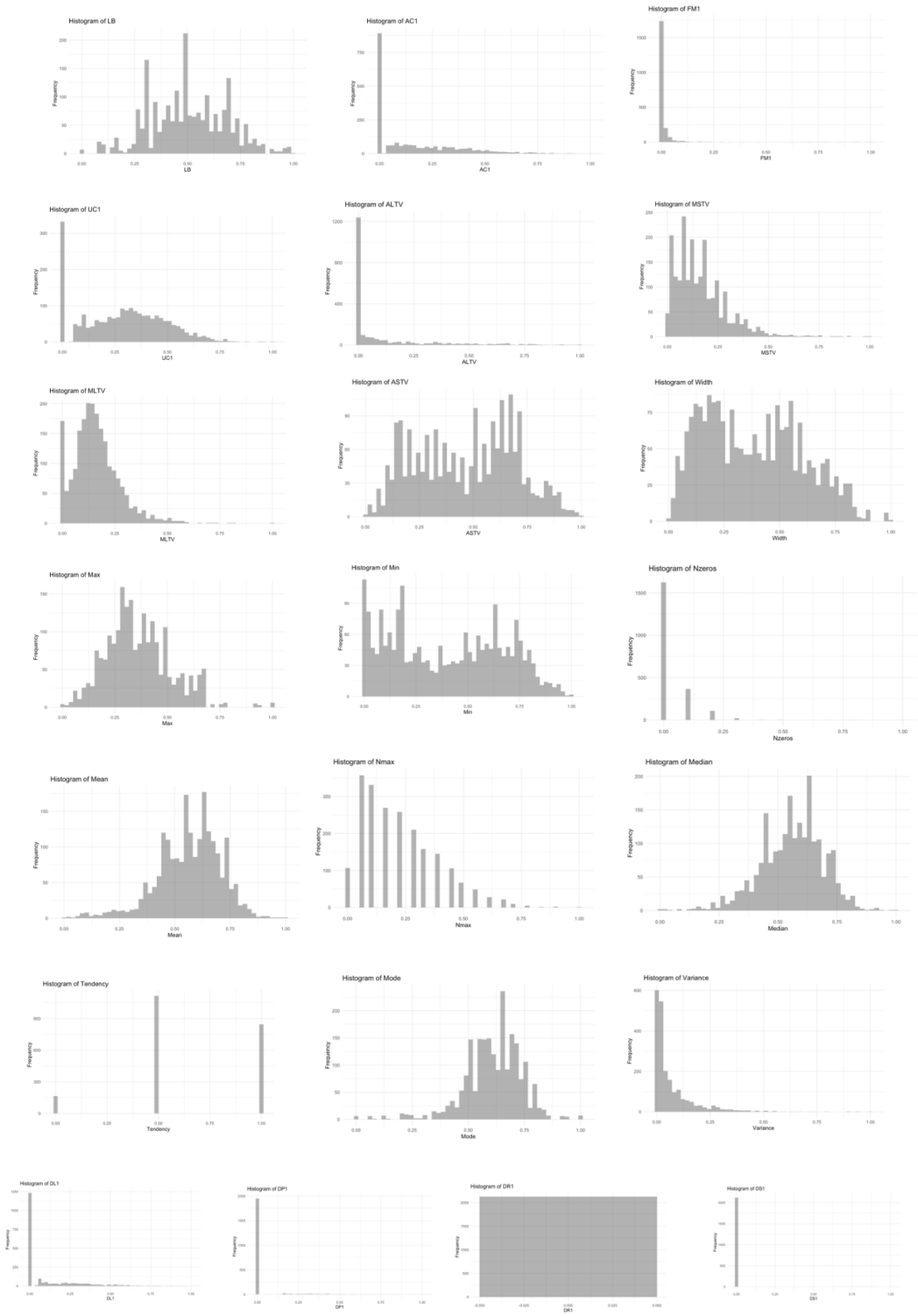

Figure 3: Histograms of the variables

Feature	Kruskal-Wallis H	df	p-value
LB	254.33	2	$< 2.2 \times 10^{-16}$
AC1	452.92	2	$< 2.2 \times 10^{-16}$
FM1	27.488	2	1.074×10^{-06}
UC1	175.73	2	$< 2.2 \times 10^{-16}$
ASTV	533.44	2	$< 2.2 \times 10^{-16}$
MSTV	383.31	2	$< 2.2 \times 10^{-16}$
ALTV	496.07	2	$< 2.2 \times 10^{-16}$
MLTV	165.62	2	$< 2.2 \times 10^{-16}$
DL1	116.05	2	$< 2.2 \times 10^{-16}$
DS1	55.521	2	8.786×10^{-13}
DP1	541.94	2	$< 2.2 \times 10^{-16}$
DR1	NaN	2	NA
Width	126.24	2	$< 2.2 \times 10^{-16}$
Min	162.49	2	$< 2.2 \times 10^{-16}$
Max	8.8154	2	0.01218
Nmax	38.77	2	3.812×10^{-09}
Nzeros	12.194	2	0.00225
Mode	268.72	2	$< 2.2 \times 10^{-16}$
Mean	302.79	2	$< 2.2 \times 10^{-16}$
Median	293.63	2	$< 2.2 \times 10^{-16}$
Variance	228.52	2	$< 2.2 \times 10^{-16}$
Tendency	71.341	2	3.224×10^{-16}

Table 2: Kruskal-Wallis Test Results: FHR Feature Differences Across Fetal States (NSP)

in Table 3.2.1, 21 of 22 explanatory variables demonstrated statistically significant distributional differences ($p < 0.05$) between groups, with most showing extreme significance ($p < 2.2 \times 10^{-16}$).

In particular, time-domain variability measures (ASTV, MSTV, ALTV, MLTV) and deceleration parameters (DP1, DL1) exhibited particularly strong discriminative power. The lone exception was DR1, which produced undefined test results (NaN) likely due to zero variance across groups, suggesting this feature should be excluded from subsequent analyses (Appaji et al., 2019).

These findings confirm that FHR characteristics contain meaningful physiological information for fetal state classification, supporting their inclusion in predictive modeling. The most discriminatory features (ASTV, DP1, ALTV) align with established clinical knowledge about pathological fetal heart patterns.

3.2.2 Mathematical Derivation of Binary Logistic Regression

To simplify the analysis, the response variable NSP will be transformed from its original three categories Normal, Suspect, Pathologic into a binary variable. `Normal` will be assigned the value 0, and the combined categories of `Suspect and Pathologic` will be assigned the value 1. Binary logistic regression will then be performed on this recoded variable. This approach offers several advantages like simplification of interpretation, reduction of computational complexity, and increases statistical power by consolidating sparse categories. Importantly, in clinical settings where time-sensitive decisions are critical, binary classification allows for faster risk stratification, enabling healthcare professionals to act more quickly when signs of fetal distress are detected.

Problem Setup

Response Variable (Y):
Let a binary outcome variable for each observation i = 1,....,n where n=2126:-

$$Y_i = \begin{cases} 0 & \text{if Normal} \\ 1 & \text{if Suspect or Pathologic (SP)} \end{cases}$$

So the goal is to model the probability that an observation corresponds to the Suspect or Pathologic category.

Predictors:

- 21 FHR explanatory variables $(X_{i1}, \ldots, X_{i21})$ selected via Kruskal-Wallis test.
- Each X_{ij} corresponds to the value of the j^{th}feature for observation i, for j=1,. . . ,21.

Sample Size:
n observations (with n_0 Normal and n_1 Suspect and Pathologic cases).

Model Specification

The logistic regression model assumes that the log-odds of an observation being in the SP category (i.e., Y=1) is a linear combination of the predictors:

$$\log\left(\frac{P(Y_i = 1|\boldsymbol{X}_i)}{P(Y_i = 0|\boldsymbol{X}_i)}\right) = \beta_0 + \sum_{j=1}^{21} \beta_j X_j$$

where β_0 is the intercept.

Equivalently:

$$P(Y_i = 1|\boldsymbol{X}_i) = \frac{1}{1 + e^{-(\beta_0 + \boldsymbol{\beta}^T \boldsymbol{X}_i)}} \quad \text{(Logistic/Sigmoid function)}$$

$$P(Y_i = 0|\boldsymbol{X}_i) = 1 - \pi_i$$

where $\boldsymbol{\beta}^T$ is the vector of regression coefficients.

Likelihood Function

For n independent observations:

$$L(\boldsymbol{\beta}) = \prod_{i=1}^{n} \pi_i^{y_i} (1 - \pi_i)^{1 - y_i}, \quad \pi_i = P(Y_i = 1|\boldsymbol{X}_i)$$

Log-Likelihood

$$\ell(\boldsymbol{\beta}) = \sum_{i=1}^{n} [y_i \log \pi_i + (1 - y_i) \log(1 - \pi_i)]$$

Substitute π_i:

$$\ell(\boldsymbol{\beta}) = \sum_{i=1}^{n} \left[y_i(\beta_0 + \boldsymbol{\beta}^T \boldsymbol{X}_i) - \log(1 + e^{\beta_0 + \boldsymbol{\beta}^T \boldsymbol{X}_i}) \right]$$

Gradient (Score Function)

First derivatives w.r.t. β_j:

$$\frac{\partial \ell}{\partial \beta_j} = \sum_{i=1}^{n} \left(y_i - \frac{e^{\beta_0 + \beta^T X_i}}{1 + e^{\beta_0 + \beta^T X_i}} \right) X_{ij} = \sum_{i=1}^{n} (y_i - \pi_i) X_{ij}$$

In matrix form:

$$\nabla \ell(\boldsymbol{\beta}) = \boldsymbol{X^T(Y - \pi)}$$

where

- $\boldsymbol{X} \in \mathbb{R}^{n \times 21}$: Design matrix of predictors (excluding intercept).
- $\boldsymbol{Y} \in \mathbb{R}^n$: Vector of responses.
- $\boldsymbol{\pi} \in \mathbb{R}^n$: Vector of predicted probabilities π_i.

Hessian Matrix

Second derivatives:

$$\frac{\partial^2 \ell}{\partial \beta_j \partial \beta_k} = -\sum_{i=1}^{n} \pi_i (1 - \pi_i) X_{ij} X_{ik}$$

Matrix form:

$$H(\boldsymbol{\beta}) = -\boldsymbol{X^T W X}$$

where $\boldsymbol{W} \in \mathbb{R}^{n \times n}$: A diagonal matrix with elements $W_{ii} = \pi_i(1 - \pi_i)$.

Newton-Raphson Optimization

Set score function to zero and solve. There is no closed form solution to score function meaning $\hat{\beta}$ needs to be determined numerically. Hence, the iterative numerical method Newton-Raphson.

Iterative updates:

$$\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} - H^{-1}(\boldsymbol{\beta}^{(t)}) \nabla \ell(\boldsymbol{\beta}^{(t)})$$

Substituted form:

$$\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} + (\boldsymbol{X}^T \boldsymbol{W}^{(t)} \boldsymbol{X})^{-1} \boldsymbol{X}^T (\boldsymbol{Y} - \boldsymbol{\pi}^{(t)})$$

Convergence is determined by monitoring the change in the parameter estimates between successive iterations. If these changes are sufficiently small, the algorithm is considered to have converged (Ratcliffe et al., 2002).

3.2.3 Binary Logistic Regression in R

Model Fitting

Using R's `glm` function (binomial family), the fitted model's estimated coefficients, standard errors, z-values, and p-values are shown in Table 3. The variable `Max` produces NA values, which could be due to two possible reasons: it is either perfectly correlated with one or more variables in the model, or it has a constant value.
To explore this, a pairwise scatterplot matrix (see Figure 4) was examined, focusing on `Max`, `Width`, `Min`, `Mean` and `Median`, which are conceptually similar summary statistics. The scatterplot showed that `Max` is almost perfectly linearly related to `Mean`, `Width` and `Median`, forming tightly clustered diagonal patterns with minimal spread. There's minimal vertical or horizontal spread in the pairwise plots, which means the variables are not independently contributing much information and they move together. This strong visual correlation implies that `Max` does not contribute unique information to the model and is likely causing redundancy among predictors. Such multicollinearity leads to instability

in coefficient estimation, which can result in inflated standard errors and convergence issues (Daoud, 2017). Given the diagnostic evidence from the pairwise scatterplot and the presence of `NA` in the regression output, the removal of `Max` is statistically sound and improves the model without significant loss of predictive power.

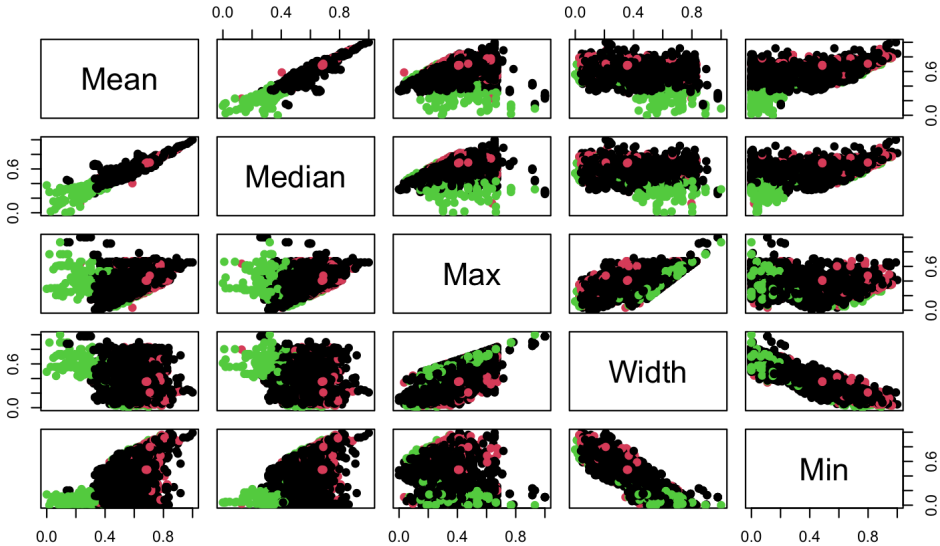

Figure 4: Pairwise Scatter Plot Matrix: Mean, Median, Max, Min, and Width

The model's AIC is 837.34. This Binary Logistic Regression (BLR) model successfully identifies several key predictors that are significantly associated with the likelihood of an observation being classified as Suspect or Pathologic. However, there is still room for improvement.

Lasso Logistic Regression Model

Although many variables in the model are not statistically significant ($p > 0.05$), removing them outright may not be the best approach. Instead, applying a regularization technique such as Lasso regression could help in selecting relevant variables while also improving model performance.

A Lasso logistic regression model was fitted using cross-validation to identify the optimal regularization parameter (λ), which was then used to refit the final model and extract the corresponding coefficients (Fonti et al., 2017). Table 4 shows the output, `DL1, Nzeros, Median` were removed from the model, suggesting they provide little to no predictive value.

A new Binary Logistic Regression (BLR) model was fitted again after excluding the variables omitted by the Lasso model and model fitting. The resulting AIC decreased slightly from 837.34 to 831.48. Although this reduction is modest, it indicates progress toward a more efficient model and suggests that further refinement may yield additional improvement.

Outliers

Examining the data for outliers is a key first step towards improving the model's efficiency. An outlier is an observation whose response value is unusual given its pattern of covariates (Zhang, 2016). Outlier can be detected using the studentized residuals. Using the `rstudent()` command studentized residuals was calculated for the fitted binary logistic regression model and then the results were visualized graphically as shown in Figure 4. Many of the observations are on zero or very close to zero whereas there are observations which are close to either sides of 2 indicating of potential outliers.

A graphical approach, using Pearson residuals, was adopted to provide a clear overview of potential outliers. Pearson residuals were calculated for the fitted binary logistic regression model with the `residuals()` function, and the visualization is shown in Figure 5.

Predictor	**Estimate**	**Std. Error**	**p-value**
Intercept	-11.13861	1.31984	$< 2e-16$
LB	-0.06499	1.49343	0.965
AC1	-16.51676	2.02771	3.78e-16
FM1	2.73666	1.15077	0.0174
UC1	-3.94832	0.54510	4.38e-13
ASTV	6.74994	0.70441	$< 2e-16$
MSTV	-1.92263	1.63553	0.2398
ALTV	2.65325	0.50854	1.81e-07
MLTV	0.65087	1.53375	0.671
DL1	0.07818	0.92665	0.933
DS1	1.61397	1.69636	0.341
DP1	14.50915	1.61916	$< 2e-16$
Width	8.19886	1.58171	2.18e-07
Min	7.53240	1.35360	2.63e-08
Max	NA	NA	NA
Nmax	1.06337	0.87086	0.222
Nzeros	-0.08090	1.27095	0.949
Mode	-8.89243	2.81271	0.00157
Mean	10.24931	2.89781	0.000405
Median	-1.29919	3.69687	0.725
Variance	9.69039	1.84494	1.50e-07
Tendency	0.94804	0.46564	0.04175

Table 3: Logistic Regression Coefficients with Standard Errors and p-values

Variable	**Coefficient**
(Intercept)	-10.0312
LB	0.1000
AC1	-14.8432
FM1	2.4099
UC1	-3.7332
ASTV	6.4379
MSTV	-1.7849
ALTV	2.5990
MLTV	0.6007
DL1	–
DS1	1.9202
DP1	13.6142
Width	6.3290
Min	5.8476
Nmax	0.9604
Nzeros	–
Mode	-7.1354
Mean	8.1795
Median	–
Variance	8.9727
Tendency	0.5489

Table 4: Lasso Logistic Regression Coefficients

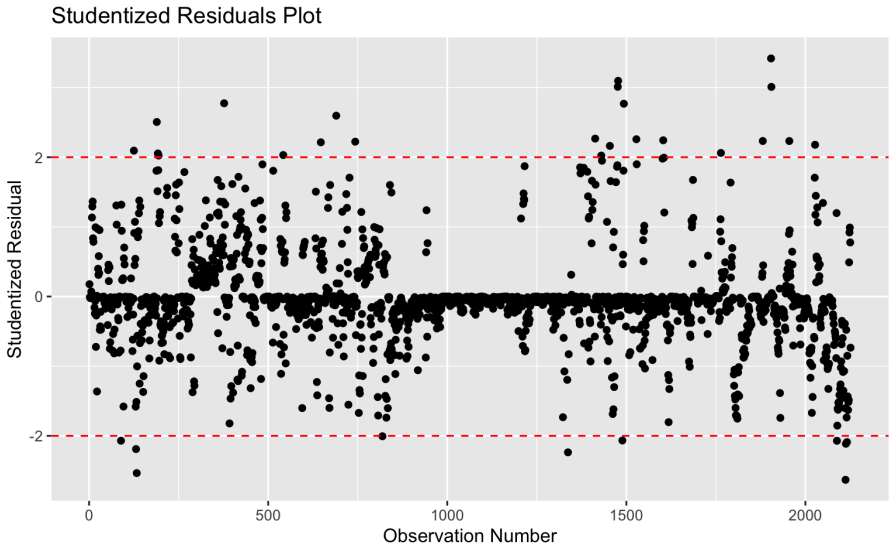

Figure 5: Studentized Residuals Plot

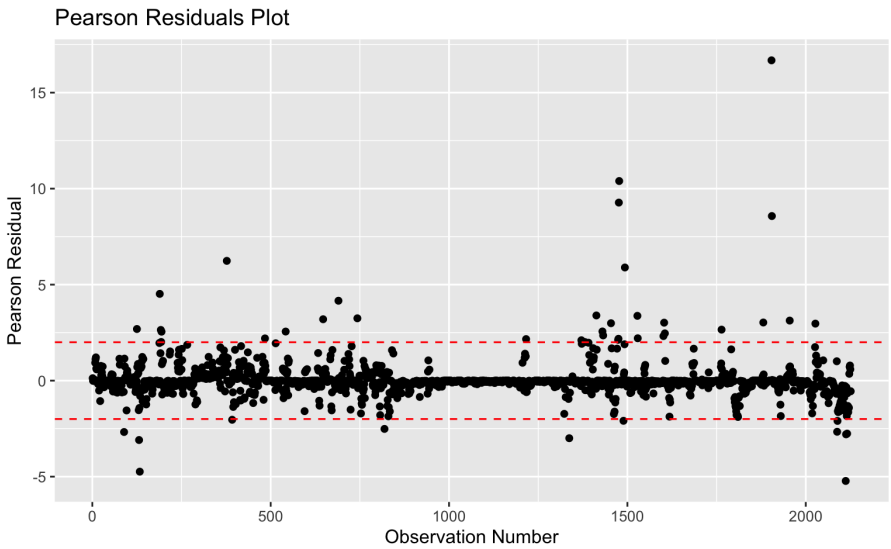

Figure 6: Pearson Residual Plot

As evident in the figure, potential outliers are present beyond the ±2 threshold, which may reduce the accuracy and introduce bias into the fitted binary logistic regression model. Consequently, the next step involves identifying and possibly removing these outliers (which will be done in the end).

Leverage

Leverage, a diagnostic concept, lacks a strict definition. Broadly, a data point distant from the majority of the others is termed a leverage point. It is important to distinguish leverage points from influential points, as the former merely have the potential to affect the model fit, but do not necessarily do so (Ellis et al., 1992). Leverage of the fitted binary logistic regression model was caluculated by using the command `hatvalues()` and the visualization is shown in Figure 6.

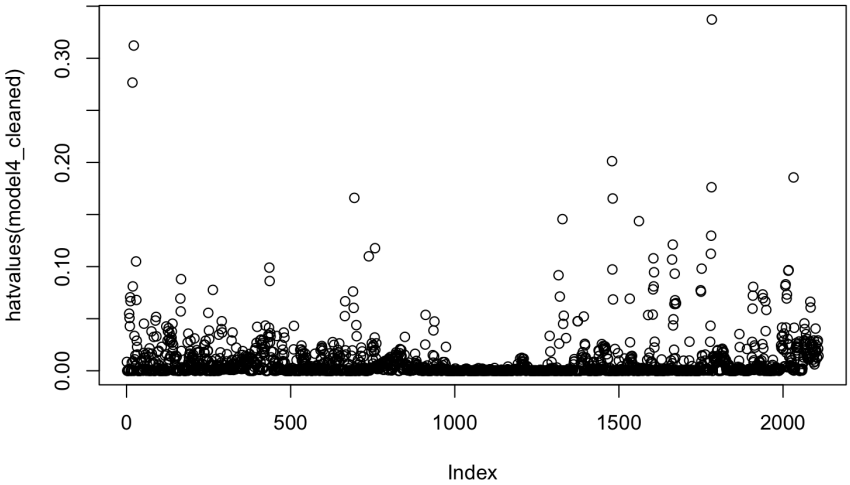

Figure 7: Hat Values (Leverage) Plot

Hat values quantify the leverage of each data point, indicating its potential to influence its own predicted value. As seen in Figure 6, most observations exhibit relatively low hat values. However, observations around index 694 and 1765 have hat values exceeding 0.2, indicating higher leverage. These points are located relatively far from the rest of the data in terms of their predictor variable values. It is important to note that high leverage does not automatically imply that these observations are influential. Further investigation using Cook's distance is needed to determine whether these high-leverage points also have large residuals.

Influence

An influential observation is one whose removal causes a substantial change in the estimated coefficients (Zhang, 2016). An influential observation is one whose removal causes a substantial change in the estimated coefficients (Zhang, 2016). Cook's distance is a summary measure of influence (Cook, 1977). Using the command `cooks.distance()`, Cook's distance was calculated for the fitted binary logistic regression model and the visualization is shown in Figure 7.

Cook's distance measures the overall influence of each data point on the fitted binary logistic regression model. As seen in Figure 7, most observations exhibit relatively small Cook's distance values. However, observation 1500 has a Cook's distance of approximately 0.06, indicating that it has a substantial influence on the model. Observation 694 also exhibits a Cook's distance above 0.02. These influential points warrant careful attention, as their inclusion in the model may be distorting the estimated relationships between the predictor variables and the response variable.

Figure 8 plot combines studentized residuals, hat values, and Cook's distance to identify influential observations. Observation 1796 stands out with a high hat value (around 0.3) and a moderately high studentized residual (around 1), resulting in a relatively large Cook's distance. This suggests that

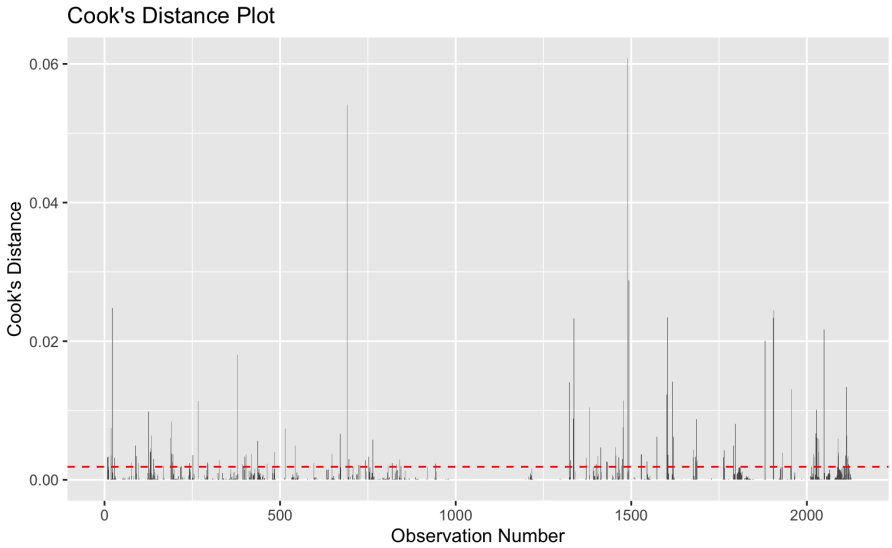

Figure 8: Cook's Distance Plot

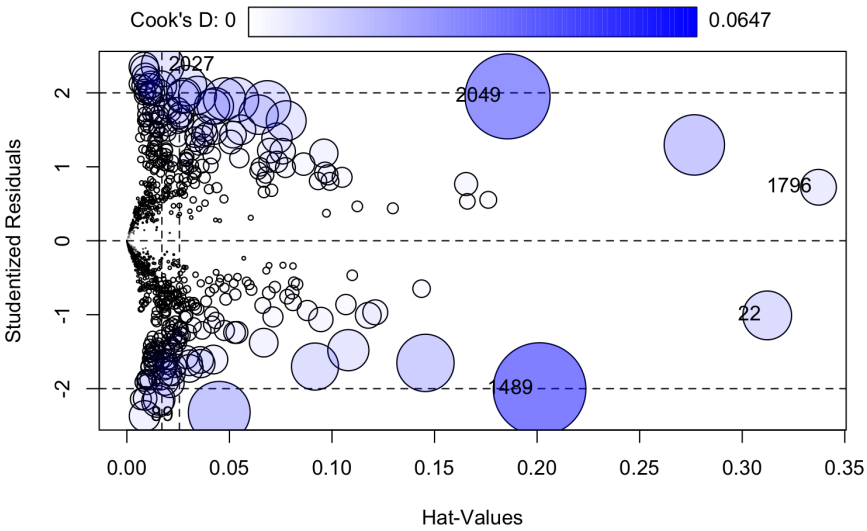

Figure 9: Studentized residuals against hat-values

observation 1796 has a considerable influence on the model. Observation 22 has a very high hat value but a low studentized residual, indicating high leverage but low influence. The size of circle is proportional to Cook's distance.

New Model

The data visualization process identified the existence of both outliers and influential observations. Fitting a new model, excluding these points, may offer advantages in terms of model performance. To identify potential outliers and influential observations within the dataset, I computed two key diagnostic metrics: Pearson residuals, and Cook's distance. For each metric, appropriate thresholds were applied—standardized thresholds for residuals which is 2 and the common rule-of-thumb $4/n$ for Cook's distance. Observations exceeding these thresholds were flagged off. The number of observations after removal of outliers and high influence was 1989. So now, new binary logistic regression model will be fitted with the reduced explanatory variables via lasso as discussed previously. The fitted model's estimated coefficients, standard errors, z-values, and p-values are shown in Table 5.

Variable	**Estimate**	**Std. Error**	**z value**	**Pr(>|z|)**
(Intercept)	-27.7566	3.1990	-8.677	$< 2 \times 10^{-16}$ ***
LB	-5.5941	2.8165	-1.986	0.0470 *
AC1	-35.1952	5.4134	-6.502	7.95×10^{-11} ***
FM1	9.6891	3.0025	3.227	0.0013 **
UC1	-8.7921	1.1292	-7.786	6.91×10^{-15} ***
ASTV	14.9519	1.6941	8.826	$< 2 \times 10^{-16}$ ***
MSTV	-8.1413	3.9355	-2.069	0.0386 *
ALTV	6.9324	0.9940	6.974	3.08×10^{-12} ***
MLTV	3.2359	3.0402	1.064	0.2872
DS1	21.5452	887.6295	0.024	0.9806
DP1	35.2362	4.3093	8.177	2.92×10^{-16} ***
Width	16.9660	3.6559	4.641	3.47×10^{-6} ***
Min	14.9227	2.9414	5.073	3.91×10^{-7} ***
Nmax	1.8915	1.7230	1.098	0.2723
Mode	-10.6997	4.7806	-2.238	0.0252 *
Mean	21.7616	5.8870	3.697	0.0002 ***
Variance	26.2704	4.0966	6.413	1.43×10^{-10} ***
Tendency	2.2798	0.8162	2.793	0.0052 **

Significance codes: $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$

Table 5: Binary Regression Results for the new model

Most predictors were statistically significant at conventional levels, including `AC1, UC1, ASTV, ALTV, DP1, Width, Min, Mean and Variance`. These variables had substantial effects on the log-odds of the outcome, with some—such as AC1, ASTV, and Variance—showing particularly strong associations ($p < 0.001$).

Certain variables, such as `DS1, Nmax and MLTV`, were not significant and may be considered for removal in future model simplifications.

The model achieved a residual deviance of 288.98 on 1971 degrees of freedom, significantly lower than the null deviance of 1968.78, indicating strong model fit. The AIC of 324.98 further supports model quality, showing a substantial improvement over earlier version which was 831.48. There is a huge drop in the AIC when outliers and high influence were removed which suggests that the old model may have been overfitted.

The non-significant variables identified in Table 5 were removed, and a new binary logistic regression model was fitted using the remaining predictors. However, this revised model resulted in an increased AIC of 338.97, compared to the original model's AIC of 324.98. This suggests that removing the non-significant variables may not be advisable, as they appear to contribute meaningfully to

the model's overall explanatory power despite their individual lack of statistical significance. These variables may be capturing interactions or confounding effects that improve model fit.

4 Conclusion

This project aimed to build a robust statistical model for the classification of fetal heart rate (FHR) recordings using binary logistic regression (BLR). The initial dataset contained three classes: `Normal`, `Suspect`, and `Pathologic`, which were recoded into a binary response variable—`Normal` (0) versus `Suspect/Pathologic` (1)—>to simplify the classification problem.

Comprehensive data preprocessing steps were undertaken to ensure model validity and enhance interpretability. Count-based variables such as the number of accelerations, fetal movements, and uterine contractions were converted into rates per second. This transformation standardised the time-dependent nature of the features, enabling fair comparisons across samples and mitigating the risk of introducing time-related bias. Additionally, variables were normalized to address scale differences, which is particularly important in regression models sensitive to magnitudes of input features.

Outliers and highly influential observations were identified using Pearson residuals, Studentized residuals, and Cook's distance. After careful evaluation, a subset of highly influential and negative outliers was excluded from the data to improve model stability and accuracy. This resulted in a substantial decrease in AIC (from 831.48 to 324.98), supporting the decision to exclude these observations.

The refined logistic regression model identified several statistically significant predictors of abnormal FHR patterns, including features such as AC1, FM1, UC1, ASTV, ALTV, DP1, Width, and Min. These variables align well with clinical expectations, as they reflect fetal heart variability, the presence of movement, and other physiological indicators.

Although a few predictors had non-significant p-values, their removal led to a higher AIC and reduced model performance, suggesting that they may still provide valuable interaction or adjustment effects.

From a clinical perspective (Hooper and Elsamadicy, 2014), one of the key challenges in interpreting fetal heart rate tracings is the variability in physician awareness and adherence to classification standards, such as those outlined by ACOG (American College of Obstetricians and Gynecologists). Even among those aware of these guidelines, the ambiguity present in Classification often leads to differing interpretations and subsequent clinical decisions. This inconsistency can result in varied fetal outcomes and influence the cost and quality of maternal and neonatal care.

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